Module code: BIO311
Module title: Modeling for Computational Biology
Semester 2, 2024/25, Year (4)

Final Examination (Resit)

Time allowed: 3 hours

Module Examiner:
Department of Biosciences and Bioinformatics, Tel.

Internal Use Only

INSTRUCTIONS to CANDIDATES

1. This is a **closed-book** examination, **which is to be written without books or notes.**
2. The total number of marks is 100. The numbers in brackets indicate the marks assigned to a section, question or sub-question.
2. The exam consists of five questions with various sub-questions. You need to answer **ALL** questions and all parts. There is no penalty for providing a wrong answer.
3. You need to motivate your answer when explicitly required.
4. Only English answers will be accepted. Answers should be written in the *Answer Booklet* provided.
5. Answers should be written using pen, black or blue, **NO PENCIL**.
6. All examination materials must be returned to the invigilator upon completion of the exam. Failure to do so will be deemed as academic misconduct and will be dealt with according to University's policy.

Question 1 [20 marks]

```
[ LEU ]
 [ atoms ]
     N    N          -0.41570     1
     H    H           0.27190     2
    CA    CT         -0.05180     3
    HA    H1          0.09220     4
    CB    CT         -0.11020     5
   HB1    HC          0.04570     6
   HB2    HC          0.04570     7
    CG    CT          0.35310     8
    HG    HC         -0.03610     9
   CD1    CT         -0.41210    10
  HD11    HC          0.10000    11
  HD12    HC          0.10000    12
  HD13    HC          0.10000    13
   CD2    CT         -0.41210    14
  HD21    HC          0.10000    15
  HD22    HC          0.10000    16
  HD23    HC          0.10000    17
     C    C           0.59730    18
     O    O          -0.56790    19
[ bonds ]
     N     H
     N    CA
    CA    HA
    CA    CB
    CA     C
    CB   HB1
    CB   HB2
    CB    CG
    CG    HG
    CG   CD1
    CG   CD2
   CD1  HD11
   CD1  HD12
   CD1  HD13
   CD2  HD21
   CD2  HD22
   CD2  HD23
     C     O
    -C     N
[ dihedrals ]
     C    CA    CB    CG        torsion_LEU_C_CA_CB_CG_mult1
     C    CA    CB    CG        torsion_LEU_C_CA_CB_CG_mult2
     C    CA    CB    CG        torsion_LEU_C_CA_CB_CG_mult3
[ impropers ]
    -C    CA     N     H
    CA    +N     C     O
```

Figure 1

Internal Use Only

Figure 1 shows how leucine topology is represented in Gromacs, using an atomistic force field (amber99sb-ildn).

a) Explain what the sections [atoms], [bonds], and [dihedrals] mean and provide a brief explanation of what the list in each of the sections represent.

[10 marks]

b) Figure 2 shows the same amino acid (leucine) represented according to a coarse-grained force field (Sirah). Explain briefly what is the difference between atomistic and coarse-grained force fields and provide examples of these differences using Figure 2.

[10 marks]

Internal Use Only

```
[ sL ]
; Coarse grained Leucine
 [ atoms ]
      GN    GN     0.13   0
      GC    GC     0.10   1
      BCG   Y1C    0.00   2
      GO    GO    -0.23   3

 [ bonds ]
      GN    GC
      GC    GO
      GO   +GN
      GC    BCG

 [ angles ]

 [ dihedrals ]

 [ impropers ]
      BCG   GC    GN    GO
```

Figure 2

Question 2 [25 marks]

Figure 3 shows the RMSD plot of the backbone of an unknown protein from simulations at different temperatures.

a) Define RMSD and explain how it is used to evaluate whether a simulation is equilibrated.

[10 marks]

b) Estimate when the different simulations in Figure 3 reach equilibrium and evaluate the equilibrium value for each simulation (estimates may slightly vary depending on interpretation of the data and justification).

[5 marks]

c) Briefly explain why the higher-temperature simulations in the example equilibrate at higher RMSD values.

[10 marks]

Internal Use Only

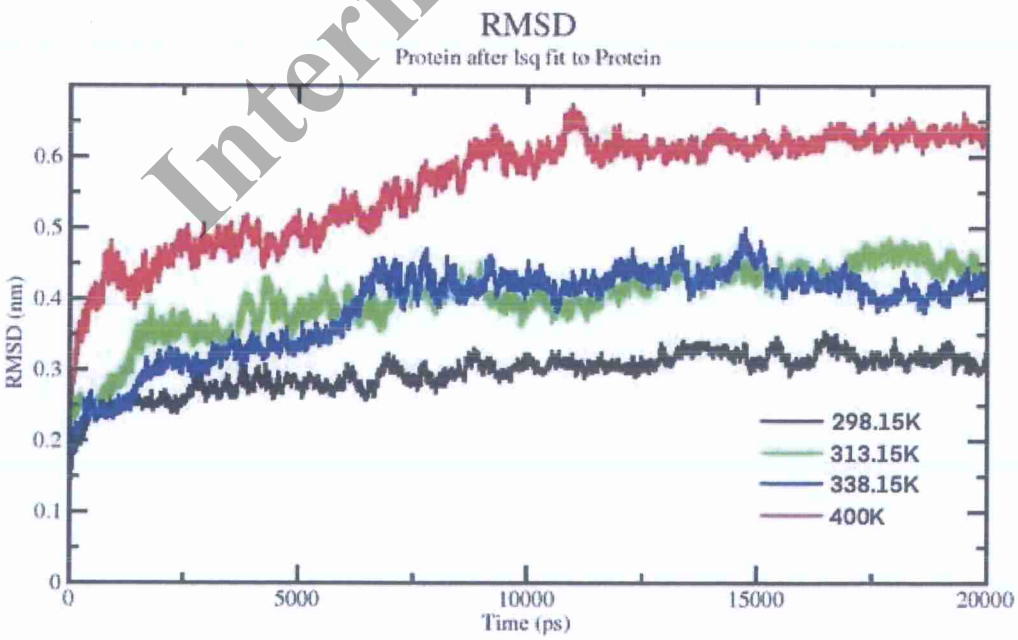


Figure 3

Question 3 [30 marks]

a) The Gibbs Free Energy is given by: $G = U + pV - TS$. Explain what the internal energy U and the entropy S are from the microscopic point of view.

[10 marks]



Figure 4

Figure 4 schematically represents the binding process of a molecule to a protein.

b) Define the binding Free Energy and explain what the difference in free energies between the bound and unbound states represents thermodynamically. Briefly explain what a negative value for ΔG means.

[10 marks]

c) Describe how molecular simulations can be used to evaluate internal energy and entropy contributions to binding free energy.

[10 marks]

Internal Use Only

Question 4 [10 marks]

Molecular Dynamics simulations in aqueous environment are often carried out using periodic boundary conditions (PBCs).

Provide a brief explanation of what periodic boundary conditions are and why they are used.

[10 marks]

Internal Use Only

Question 5 [15 marks]

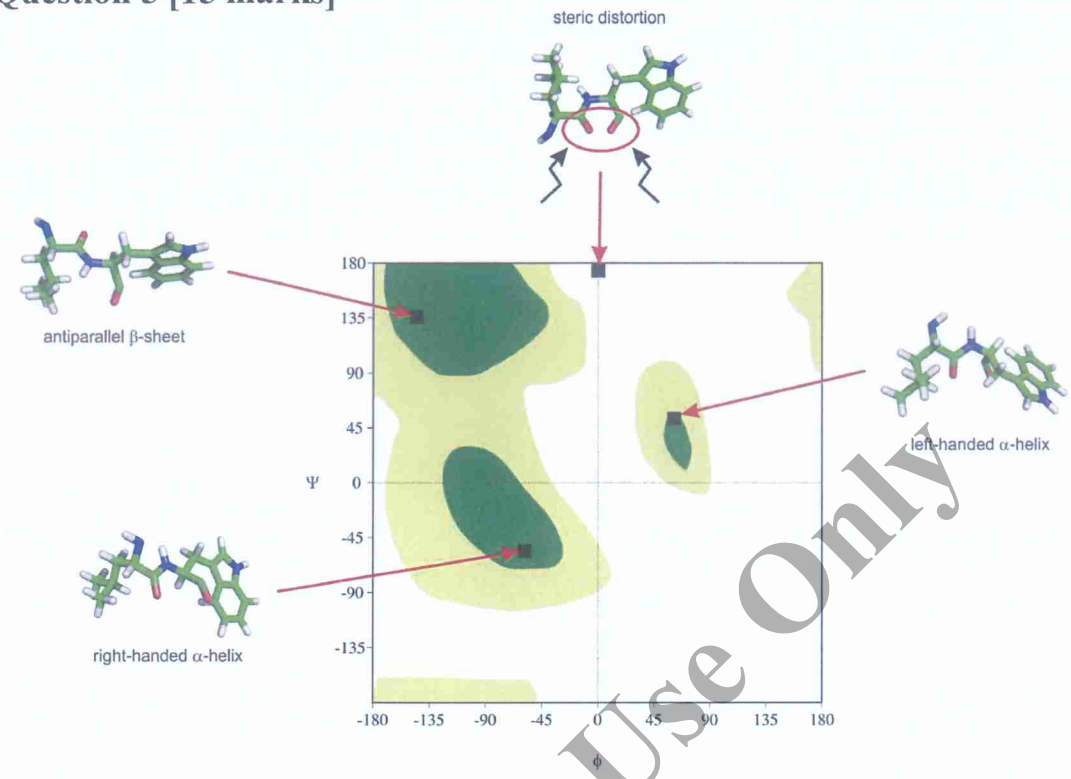


Figure 5

Figure 5 shows the typical protein Ramachandran plots.

a) Provide a brief explanation of what the two angles (φ and ψ) in the plot are.

[5 marks]

b) Write a short caption for Figure 5, explaining in particular what the light green and dark green regions represent.

[10 marks]

Internal Use Only